

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

Zheng Huang · Diego Gutiérrez

Micro- & Macroprudential Management · Country implementation of a CCyB increase

Master of Financial Technology · Frankfurt School of Finance & Management

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SAY

- Good morning. We are Zheng Huang and Diego Gutiérrez.
- Our case is the Netherlands: why the Dutch central bank, DNB, raised its countercyclical capital buffer from zero to two percent.

NUMBERS

- CCyB 0% → 2% (fully binding 31 May 2024)

TERMS EXPLAINED

- **DNB:** De Nederlandsche Bank, the Dutch central bank and macroprudential authority.

In 2022–23 the Netherlands raised its countercyclical capital buffer (CCyB) to 2% while the rulebook said 0%. It was refilling a buffer, not fighting a boom

SUMMARY

Decision: CCyB 0% → 1% (ann. May 2022) → **2%** (ann. May 2023, binding 31 May 2024), together with lower buffers for systemically important banks (O-SII).

Puzzle: the Basel reference indicator, the credit-to-GDP gap (credit above its long-run trend), stood at **−33pp** and **−47pp** at the two decisions → the buffer guide implied **0%**.

Q1 · Environment

Recovered economy, then war, energy shock and 11.6% inflation; ECB hikes from Jul-2022.

Credit cold; housing hot (+19% y/y), then falling.

Banks strong (core capital ratio, CET1, 17.7%) and more profitable as rates rose.

Q2 · Criteria & risks

DNB framework (2022): 2% = “**standard risk environment**”, calibrated on past crisis losses (€12bn), not on credit growth.

Insures against shocks no indicator predicts; housing left to other tools.

Keeps the March-2020 promise to restore releasable capital.

Q3 · Implications

Cheap: ≤ about one fifth of big banks’ CET1 headroom.

A swap: structural buffers cut in 2020 and 2024.

No sign of contraction in credit or its price.

Complements ECB tightening; reciprocated abroad; **€6.7bn** releasable.

Verdict: **normalisation, not tightening.** **Trade-off:** relies on discretion and credibility; release never tested.

Source: DNB (2020, 2022, 2023, 2026); DNB Financial Stability Reports 2022–23; BIS credit-to-GDP gaps; ECB; bank disclosures; own calculations.

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- DNB raised the buffer twice, to two percent, while the Basel indicator, the credit-to-GDP gap, pointed to **zero**.
- The three boxes answer the three questions of the assignment: environment, criteria and risks, implications.
- Our verdict: **normalisation, not tightening**. DNB refilled a buffer; it did not fight a boom. It cost banks little, was largely offset by cuts elsewhere, and we see no sign of a contraction in lending.

NUMBERS

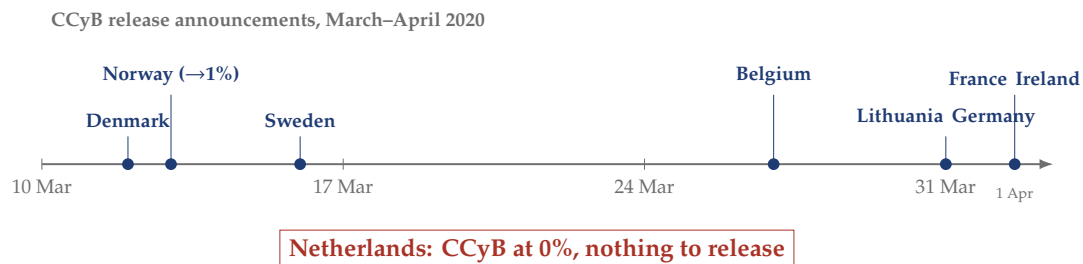
- Announcements: May 2022 (1%) and May 2023 (2%); binding 25 May 2023 and 31 May 2024
- Credit-to-GDP gap at the decisions: **−33pp** and **−47pp** → Basel guide 0%
- Inflation 11.6% (2022) · CET1 17.7% (end-2021) · **€6.7bn** releasable

TERMS EXPLAINED

- **CCyB:** Countercyclical capital buffer. Extra bank capital (0–2.5% of risk-weighted assets) built in good times, released in bad times.
- **Credit-to-GDP gap:** How far credit/GDP is above its long-run trend; the Basel starting point for the CCyB.
- **O-SII:** “Other systemically important institution”. The O-SII buffer is an extra capital charge on large banks.

In March 2020 COVID struck and neighbours released their buffers, but the Dutch buffer was empty

PROLOGUE



DNB improvised (17 Mar 2020):

- cut systemic buffers of ING, Rabobank, ABN AMRO: 3% → 2.5 / 2 / 1.5%
- postponed a planned floor on mortgage risk weights
- €8bn freed, up to €200bn of lending

... and made a promise

"Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%."

Source: ESRB, CCyB rates table (release announcements, Mar–Apr 2020); DNB press release "DNB lowers bank buffer requirements to support lending", 17 Mar 2020.

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SAY

- March 2020: COVID hits. Within three weeks our neighbours **release** their countercyclical buffers so that banks can keep lending.
- Sweden and Norway were at 2.5 percent; Denmark, Ireland and Lithuania at one percent.
- The Netherlands was at **zero**. There was nothing to release.
- France and Ireland followed on the first of April, and Belgium and Germany cancelled increases they had already announced. Across Europe, buffers built in good times were being used exactly as intended.
- So DNB improvised. It cut the systemic buffers of ING, Rabobank and ABN AMRO and postponed a mortgage rule, freeing about eight billion euros.
- DNB said this capital could support up to 200 billion euros of lending. But it was a workaround: these buffers were never designed to be released, so the cut had to be negotiated and explained bank by bank.
- And it made a promise (*pause*): once things were normal, it would rebuild this capital as a two-percent countercyclical buffer.

NUMBERS

- Releases (rate in force → new rate):
 - Denmark 1.0% → 0% (12 Mar)
 - Norway 2.5% → 1.0% (13 Mar)
 - Sweden 2.5% → 0% (16 Mar)
 - Lithuania 1.0% → 0% (31 Mar)
 - France 0.25% → 0% (1 Apr)
 - Ireland 1.0% → 0% (1 Apr)
- Cancelled pending increases: Belgium 0.5% (27 Mar) · Germany 0.25% (31 Mar)
- Also: Iceland 2.0% → 0% · Czech Republic 1.75% → 1.0%
- NL: systemic buffers 3% → 2.5% (ING) / 2% (Rabo) / 1.5% (ABN AMRO); €8bn freed (incl. the floor postponement); up to €200bn of lending

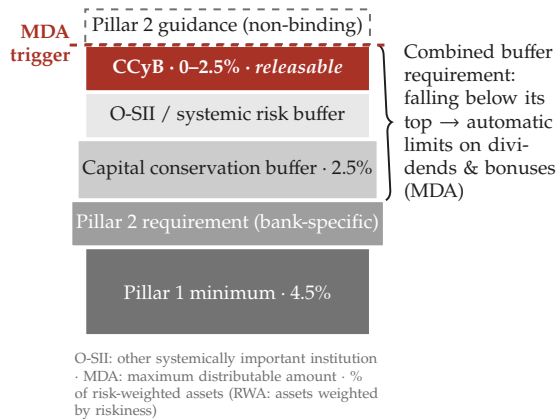
TERMS EXPLAINED

- **Release:** The authority lowers the CCyB rate with immediate effect, so banks may use that capital to absorb losses and keep lending.
- **Systemic (risk) buffer:** A structural capital charge on large banks; not designed to be released.
- **Mortgage risk-weight floor:** A minimum average risk weight on Dutch mortgages for banks using internal models.

NEXT → "That empty buffer mattered, because not all capital can be released."

That mattered because the CCyB is the only layer of bank capital an authority can release in a crisis

PROLOGUE



Design

- 0–2.5% of risk-weighted assets, on *domestic* exposures
- Increases bind after 12 months; releases are immediate
- EU reciprocity: foreign banks lending in NL hold it too

Why “releasable” matters

- In COVID, banks close to the MDA trigger **cut lending** even when *allowed* to use buffers (stigma, payout limits)
- Only an explicitly **released** buffer is truly usable

Source: CRD (Arts. 130–136); BCBS (2024) d585; Couaillier, Lo Duca, Reghezza & Rodriguez d’Acri (2022), ECB WP 2644. Stack not to scale.

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SAY

- Read the stack from the bottom: first the requirements banks must always meet, then the buffers on top.
- The red dashed line is the MDA trigger. Below it, dividends and bonuses are automatically restricted. So banks avoid using buffers even when they are allowed to; in COVID, banks near that line cut lending instead.
- The CCyB is different. The authority can switch it off, so the requirement itself falls and there is no stigma. It is the only truly **releasable** layer.
- A release is also immediate, while an increase only binds after twelve months. That asymmetry is why the buffer has to be filled before the crisis, not during it.

NUMBERS

- Pillar 1 minimum 4.5% CET1 · capital conservation buffer 2.5%
- CCyB 0–2.5% of RWA (higher possible) · increases bind after 12 months · releases are immediate

TERMS EXPLAINED

- **RWA:** Risk-weighted assets. Assets weighted by riskiness (a mortgage counts less than a corporate loan).
- **CET1:** Common Equity Tier 1. The highest-quality capital (shares, retained earnings), as % of RWA.
- **MDA:** Maximum distributable amount. The cap on payouts that applies once capital falls below the combined buffer requirement.
- **Pillar 2 requirement / guidance:** The supervisor’s bank-specific add-on (binding) / extra expectation (non-binding).

NEXT → “Two years later DNB refilled exactly that layer, at a strange moment.”

Two years later DNB refilled it in the middle of a storm and against its own rulebook. Was that the wrong moment?

THE QUESTION

The storm (2022–23)

War in Ukraine · record energy prices
 HICP inflation **11.6%** (2022)
 ECB hikes from **21 Jul 2022**
 House prices turning down
 Basel gap **–33 / –47pp** → guide **0%**

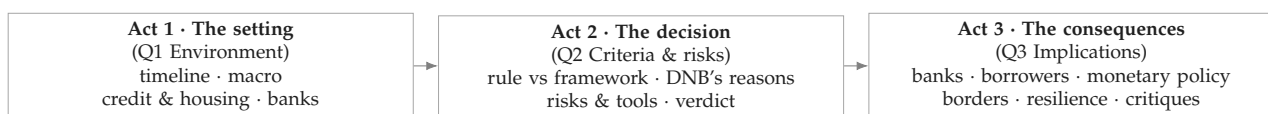


The decision

CCyB **0% → 1% → 2%**
 announced May 2022 and May 2023
 binding May 2023 and May 2024

Raising capital in a downturn amplifies the cycle. That **procyclical mistake** is what the CCyB was invented to avoid.

Tightening into a slowdown? Fighting a hidden boom? Or something else?



Source: CBS (HICP 2022); ECB monetary policy decision, 21 Jul 2022; BIS credit-to-GDP gaps (latest data at each decision: 2021Q4, 2022Q4).

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SAY

- Now look at 2022: war, record energy prices, 11.6 percent inflation, ECB rate hikes and a housing market about to turn. And the Basel indicator says zero.
- Under the Basel guide, a gap below two points means a zero buffer, and the gap was more than thirty points below that threshold.
- Raising capital in a downturn amplifies the cycle, which is exactly what the CCyB is meant to avoid.
- So our question: **was this the wrong moment?** We answer it in three acts, which are the three questions of the assignment.

NUMBERS

- HICP inflation 11.6% (2022) · ECB first hike 21 Jul 2022 (+50bp)
- Credit-to-GDP gap –33pp (data to 2021Q4) and –47pp (data to 2022Q4)

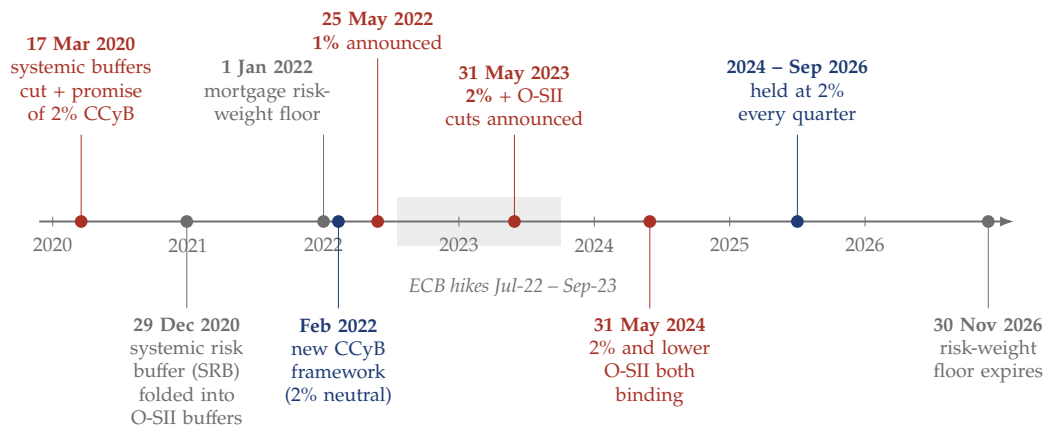
TERMS EXPLAINED

- **Procyclical:** Policy that reinforces the cycle (tighter in bad times, looser in good times).
- **Basel buffer guide:** Gap below 2pp → 0% buffer; above 10pp → 2.5%; linear in between.

NEXT → "Act 1, the setting. The refill followed a path promised in 2020."

The refill followed the path promised in 2020, one step per year, just as the ECB started to tighten

Q1 · ENVIRONMENT



Source: DNB press releases (17 Mar 2020; 27 May 2022; 31 May 2023; 17 Sep 2026; 24 Apr 2026); ESRB O-SII notifications (27 Nov 2020; 31 May 2023); ECB (21 Jul 2022; 14 Sep 2023).

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SAY

- After the promise in 2020, DNB published its framework in February 2022 and announced one percent in May 2022, just before the ECB began to hike.
- In May 2023 it announced two percent, together with lower buffers for the largest banks. Both became binding on 31 May 2024.
- It has held two percent every quarter since. The mortgage floor expires in November 2026.

NUMBERS

- 17 Mar 2020 · 29 Dec 2020 · 1 Jan 2022 · Feb 2022 · 25 May 2022 · 31 May 2023 · 31 May 2024 · 30 Nov 2026
- ECB hikes: first on 21 Jul 2022; still hiking on 14 Sep 2023

TERMS EXPLAINED

- **Announcement vs binding date:** Banks get 12 months between a CCyB increase being announced and having to meet it.
- **CRD V:** The 2019 revision of the EU Capital Requirements Directive; it made the O-SII and systemic risk buffers additive, prompting DNB's conversion.

NEXT → "By then the economy had fully recovered, but the outlook was darkening."

By then the economy had fully recovered, but war, energy prices and inflation were darkening the outlook

Q1 · ENVIRONMENT

2021Q3

GDP back above
pre-COVID level

11.6%

HICP inflation, 2022
(national CPI 10.0%)

+50bp

ECB's first hike
21 July 2022

record

energy & commodity prices
after the invasion

DNB, Financial Stability Report, spring 2022

"After a strong recovery from the pandemic, the economic outlook has worsened due to the war in Ukraine and high inflation, and financial stability risks have increased."

"Higher funding costs may put pressure on the debt sustainability of governments, businesses and households."

By end-2021 the Dutch economy was almost 3% larger than at end-2019. DNB's framework requires this recovery before the buffer is built.

Source: CBS (GDP, Nov 2021; inflation 2022); ECB monetary policy decision 21 Jul 2022; DNB Financial Stability Report, spring 2022.

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SAY

- By the third quarter of 2021 GDP was back above its pre-COVID level. That recovery is what DNB's framework requires before building the buffer.
- Then came the storm: war in Ukraine, record energy prices, inflation of 11.6 percent and the ECB's first hike in July 2022. DNB warned that higher funding costs would strain debt sustainability.
- So the macro picture argued **both ways**: build while the economy can take it, or wait because it is weakening.

NUMBERS

- GDP above pre-COVID level: 2021Q3 (+1.9% q/q) · end-2021 ≈ +3% vs end-2019
- HICP 11.6% in 2022 (national CPI 10.0%)
- ECB +50bp on 21 Jul 2022

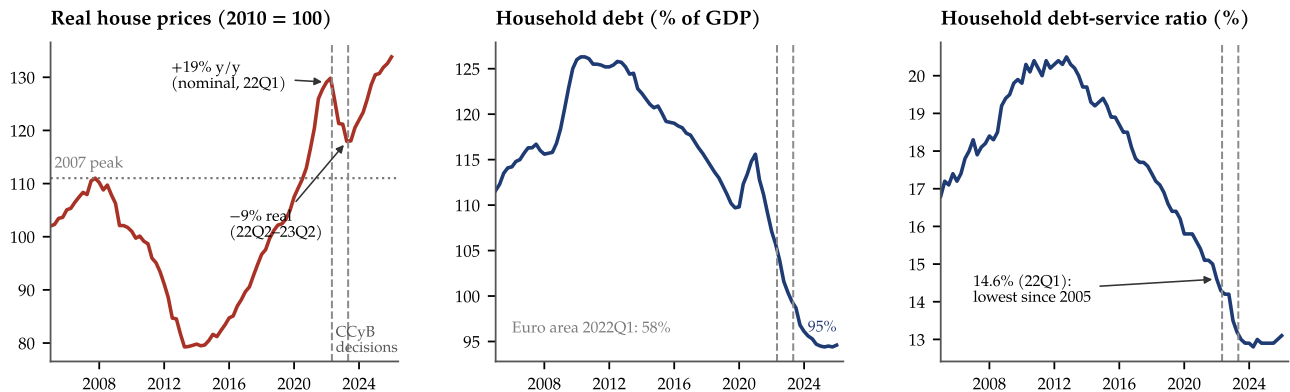
TERMS EXPLAINED

- **HICP**: Harmonised Index of Consumer Prices, the euro-area-comparable inflation measure.
- **Basis point (bp)**: 0.01 percentage point; 50bp = 0.5pp.

NEXT → "In the financial system, credit was cold and only housing was hot."

In the financial system credit was cold and only housing was hot, and even housing soon turned

Q1 · ENVIRONMENT



Housing: nominal prices +19% y/y (22Q1), real 16% above the 2007 peak. DNB itself: “the overheated housing market calls for further measures.”

Credit: household debt high (107% of GDP vs 58% in the euro area) but *falling*; debt service at its lowest since 2005.

Source: BIS (WS_SPP, WS_TC, WS_DSR); DNB Financial Stability Report, spring 2022; own calculations. Gold lines: CCyB announcements (May-22, May-23).

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SAY

- Left: **housing was hot**. Nominal prices rose 19 percent in a year, and DNB itself called the market overheated. But as mortgage rates rose, real prices fell about nine percent.
- In real terms prices were 16 percent above their 2007 peak, so the level was high even after correcting for inflation.
- Middle: household debt is high, 107 percent of GDP against 58 in the euro area, but it was **falling**.
- Right: the share of income spent on debt service was the lowest since 2005.
- If there was a boom, it was in house prices, not in credit.

NUMBERS

- Nominal house prices +19.0% y/y (2022Q1); real index 129 vs 2007 peak 111 (+16%)
- Real house prices –9.2% (2022Q2–2023Q2)
- Household debt/GDP: 110% (2019Q4) → 107% (2022Q1) → 100% (2023Q1); euro area 57.7% (2022Q1)
- Debt-service ratio, households: 14.6% (2022Q1)

TERMS EXPLAINED

- **Debt-service ratio:** Interest plus principal payments as a share of income.
- **Real house prices:** Nominal prices deflated by consumer prices.

NEXT → “And the banks entered this storm well capitalised.”

Banks entered this storm well capitalised, and rising rates were lifting their profits

Q1 · ENVIRONMENT

17.7%

core capital (CET1) ratio
end-2021 (EU 15.7%, 21Q3)

16.3%

CET1 ratio, end-2022
"in line with EU average"

+6.7%

net interest income 2022
(67.7% of income)

11.5%

4 major banks' CET1 after a
severe stress test (min. 8%)

Q1 in one line: against raising now

war and energy shock · 11.6% inflation
rapid ECB tightening · house prices turning

DNB (FSR 2022): higher rates "may ultimately have a positive impact on the profitability of financial institutions."

... and for raising now

economy recovered · credit cold
banks strong, profits rising with rates

Source: DNB Financial Stability Reports, spring 2022 and spring 2023.

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SAY

- Core capital was 17.7 percent of risk-weighted assets at the end of 2021, above the EU average, and in DNB's stress test the big banks stayed well above the minimum.
- Rising rates also **lifted profits**: net interest income grew 6.7 percent in 2022. Remember that, because it comes back.
- Act 1 in one line: war, inflation, rate hikes and a turning housing market argued against raising now; a recovered economy, cold credit and strong banks argued for it.

NUMBERS

- CET1 17.7% (end-2021; EU 15.7% in 2021Q3) → 16.3% (end-2022, in line with EU)
- Stress test (FSR 2023): -3.8pp → 11.5% at end-2025 (min. 8%)
- Net interest income +6.7% in 2022; 67.7% of bank income

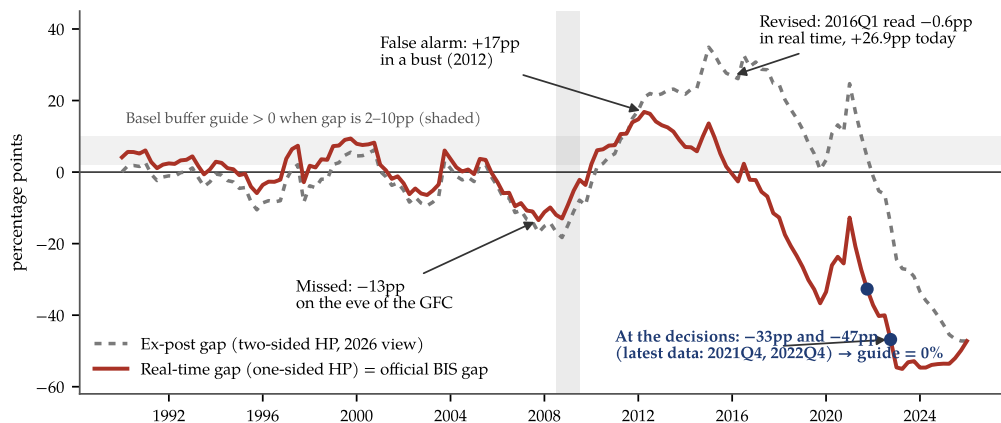
TERMS EXPLAINED

- **Stress test**: A simulation of a severe recession to check whether capital stays above the minimum.
- **Net interest income**: Interest earned on loans minus interest paid on deposits and funding.

NEXT → **HAND-OVER**: "That was the setting. So why did DNB ignore its own rulebook? Diego."

So why ignore the rulebook? On Dutch data the Basel rule had missed the last crisis and raised false alarms

Q2 · CRITERIA & RISKS



Credit-to-GDP gap = credit/GDP ratio minus its long-run trend (Hodrick-Prescott filter). **Rule:** gap < 2pp → 0%, gap > 10pp → 2.5%. Our one-sided filter ($\lambda = 400,000$) reproduces the official BIS gap within 0.00005pp (1971–2026).

Alternative trend filters are noisy around COVID (two-sided HP +4.2pp in 2021Q4), which is one more reason not to rely on any of them.

Source: BIS credit-to-GDP gaps (Q.NL.P.A); Drehmann et al. (2010); Repullo & Saurina (2011); Edge & Meisenzahl (2011); own calculations.

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SAY

- Thank you. So why ignore the rulebook? Because on Dutch data it has a poor record.
- The red line is the credit-to-GDP gap, credit relative to its long-run trend. We rebuilt it and match the official BIS series.
- Under the Basel guide the buffer starts when the gap exceeds two points and reaches its maximum at ten. So everything depends on this one number being reliable.
- It **missed** the financial crisis, at minus 13 in 2007. It raised a **false alarm** in 2012, during a bust. And it gets **revised**: 2016 read slightly negative at the time and plus 27 today.
- The reason: Dutch credit grew from about 50 to 350 percent of GDP, and the filter mistakes that structural rise for trend.
- One-sided means the filter only uses data available at each date, which is what a policymaker actually sees. That is why the signal can change years later, once new data arrive.
- At the two decisions it read minus 33 and minus 47. The rule said zero.

NUMBERS

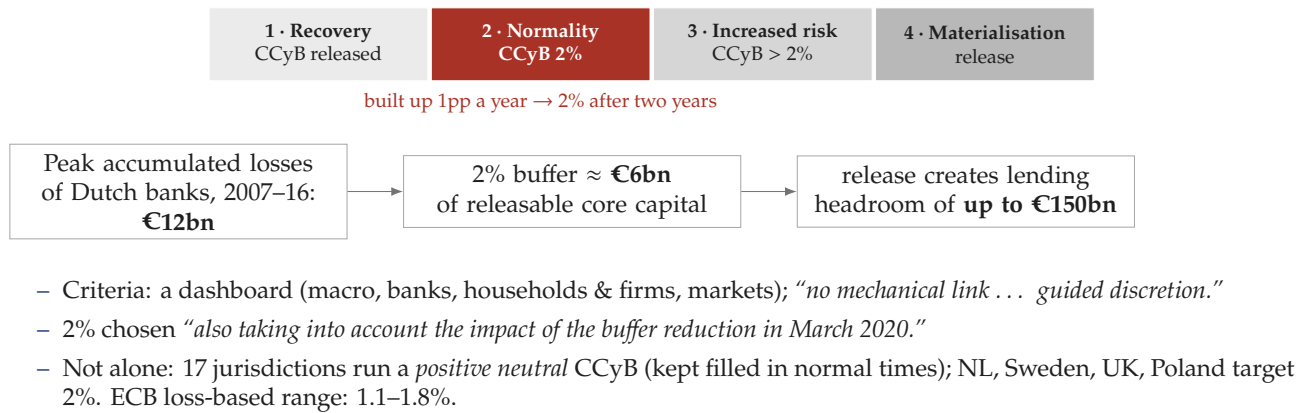
- Gap: −13.4pp (2007Q4) · +16.8pp (2012Q2) · 2016Q1 −0.6pp real time vs +26.9pp ex post
- −32.7pp (2021Q4) and −46.8pp (2022Q4) at the decisions
- Credit/GDP 47% (1961) → 352% (2015Q1 peak)
- Replication error ≤ 0.00005 pp (1971–2026)

TERMS EXPLAINED

- **HP filter:** Splits a series into a smooth trend and a cycle; $\lambda = 400,000$ for credit cycles.
- **One-sided vs two-sided filter:** One-sided uses only data up to each date (what policymakers see); two-sided also uses later data (hindsight).
- **False alarm / missed crisis:** Type II / type I errors of an early-warning indicator.

DNB used its own compass instead: 2% in normal times, calibrated on past crisis losses rather than on credit growth

Q2 · CRITERIA & RISKS



Source: DNB (2022), Analytical framework for setting the CCyB in the Netherlands; BCBS (2024) d585, Table 1; De Nora, Pereira, Pirovano & Stammwitz (2025), ECB WP 3061.

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SAY

- DNB's 2022 framework has four phases: release after a crisis, **two percent in normal times**, more when risks are high, and release again in a crisis.
- This is a *positive neutral* buffer: already filled when risks are neither high nor low.
- The two percent comes from history. Past crises cost Dutch banks about 12 billion euros at the peak. Two percent is about six billion of releasable capital, enough to support up to 150 billion of lending.
- The ECB's own loss-based estimates give 1.1 to 1.8 percent, so two percent is at the top of that range but not outside it. We come back to that in the critiques.
- Seventeen jurisdictions now work this way; Sweden, the UK and Poland also target two percent.

NUMBERS

- Peak accumulated losses €12bn (2007–2016) → 2% ≈ €6bn → up to €150bn of lending
- Build-up: 1pp per year → 2% after two years
- 17 jurisdictions with a positive neutral CCyB (BCBS 2024); ECB loss-based range 1.1–1.8%

TERMS EXPLAINED

- **Positive neutral CCyB:** A CCyB kept above zero in a "standard" risk environment.
- **Peak accumulated losses:** The largest cumulative loss a bank suffered over a crisis period.
- **Guided discretion:** Decisions informed by a published set of indicators, with judgement, not a formula.

NEXT → "And its stated reasons were never credit growth."

Its stated reasons were recovery, uncertainty and the 2020 promise, never credit growth

Q2 · CRITERIA & RISKS

May 2022 · 0% → 1%

*"When the systemic risk buffers . . . were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB."*

"The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but . . . the robust economic recovery . . . provides grounds for a gradual build-up."

Cyclical risks: "normal" to elevated.

May 2023 · 1% → 2%

*"Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk."*

Cited: investors' rising risk appetite · real-estate prices falling · debt sustainability of firms and governments · banks "robust, partly due to rising interest rates".

Both decisions: **a normal-to-elevated risk picture + strong banks + the promise**. Credit growth is never the reason.

Source: DNB Financial Stability Report, spring 2022; DNB press releases of 27 May 2022 and 31 May 2023.

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SAY

- In 2022 DNB wrote that it was **restoring the buffers** it had lowered in March 2020, citing the recovery and a normal-to-elevated risk picture.
- In 2023 it said the credit gap showed **no signs of excessive credit growth**, and pointed instead to risk appetite, falling property prices, debt sustainability and strong banks.
- Credit growth was never the reason.

NUMBERS

- 1% step: about €3.3bn of extra CET1 (announced 25/27 May 2022)
- 2% step: about €3.4bn (announced 31 May 2023)

TERMS EXPLAINED

- **Cyclical systemic risk:** Risk that builds up over the financial cycle (credit, asset prices, risk-taking) and can materialise system-wide.

NEXT → "So what was the buffer meant for?"

The buffer was meant for shocks nobody can forecast, while the known hot spots got their own tools

Q2 · CRITERIA & RISKS

Financial-stability risks named by DNB (FSR 2022/23)	Instrument
War, energy and supply shocks · rapid tightening of financial conditions and market corrections · debt sustainability of governments, firms, households · bank turmoil abroad (SVB, Credit Suisse, 2023)	CCyB, releasable for <i>any</i> shock
"Overheated" housing · high mortgage debt of first-time buyers · interest-only loans	borrower limits (loan-to-value, loan-to-income), tax reform, mortgage risk-weight floor (Jan 2022 – Nov 2026)
Size and concentration of the large banks (sector 400% → 280% of GDP)	buffer for systemically important banks (O-SII , cut 2024)

Principle: one instrument per target (Tinbergen, 1952)

DNB on the 2020 recomposition: more releasable capital is "*a valuable addition given the sensitivity/volatility of the Dutch economy to external events.*" A releasable buffer is the only tool that works whatever the source of the shock.

Source: DNB Financial Stability Reports, spring 2022 and spring 2023; ESRB O-SII notification NL (31 May 2023); DNB (24 Apr 2026); Tinbergen (1952).

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SAY

- Each risk got its own tool.
- This matters for our question. If housing had been the target, the CCyB would be the wrong tool: it hits all lending, not just mortgages.
- **Shocks nobody can forecast**, such as war, an energy shock, a sudden tightening or bank turmoil abroad, go to the CCyB, because it can be released whatever the source.
- **Housing**, the known hot spot, went to borrower limits, tax reform and a mortgage risk-weight floor. The **size of the big banks** goes to the systemic buffer.
- One instrument per target: the Tinbergen principle.

NUMBERS

- Banking sector: $\approx 400\%$ of GDP (when O-SII buffers were phased in) → 280% (end-2022)
- Risk-weight floor in force Jan 2022 – Nov 2026

TERMS EXPLAINED

- **LTV / LTI:** Loan-to-value / loan-to-income. Caps on mortgage size relative to house value or income.
- **Tinbergen principle:** Reach each policy target with its own instrument.
- **Borrower-based measures:** Rules on borrowers (LTV, LTI) rather than on bank capital.

NEXT → "Now we can test the evidence against both readings."

Tested against the evidence, every prediction fits normalisation and none fits overheating

Q2 · CRITERIA & RISKS

If DNB were ...	normalising	fighting overheating	what happened	
Responding to credit indicators	no link	rises with them	gap, debt service, household debt ratio all <i>falling</i> ; CCyB <i>up</i>	✓
Path	+1pp a year, stop at 2%	follows risk	1% → 2% → held	✓
2023: house prices falling, rates jumping	raise anyway	pause	raised	✓
2024–26: house prices rising again (+10% y/y, Mar-26)	stay at 2%	go above 2% (framework phase 3)	held at 2%	✓
Other buffers	cut structural ones to compensate	keep them	cut in 2020 and 2024	✓

Five predictions, five matches: **normalisation**. The last row is the key to the whole story.

Source: BIS (WS_CREDIT_GAP, WS_DSR, WS_TC, WS_SPP); DNB press releases 2022–2026 (house prices: DNB, 26 Mar 2026); DNB framework (2022); ESRB O-SII notifications.

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SAY

- Overheating or normalisation? Each predicts something different. Row by row:
- If DNB was fighting a boom, the buffer should follow credit and prices. If it was refilling, it should follow a fixed path back to its normal level. These readings make different predictions, so we can test them.
- **Indicators.** Overheating says the buffer rises with them. They all fell while the buffer rose.
- **Path.** Normalisation says one point a year, then stop at two. That is exactly what happened.
- **2023.** Prices falling, rates jumping. An overheating fighter pauses; DNB raised.
- **2024 to 2026.** Prices up ten percent again. An overheating fighter goes above two; DNB held.
- **Other buffers.** Normalising means cutting structural buffers in return. DNB did, twice.
- (*pause*) Five out of five.
- Every observation fits normalisation, and none fits overheating. This is the core of our argument.

NUMBERS

- Gap, debt-service ratio and household debt ratio all falling in 2022–23
- House prices +10% y/y (DNB, Mar 2026) · CCyB held at 2% since May 2024

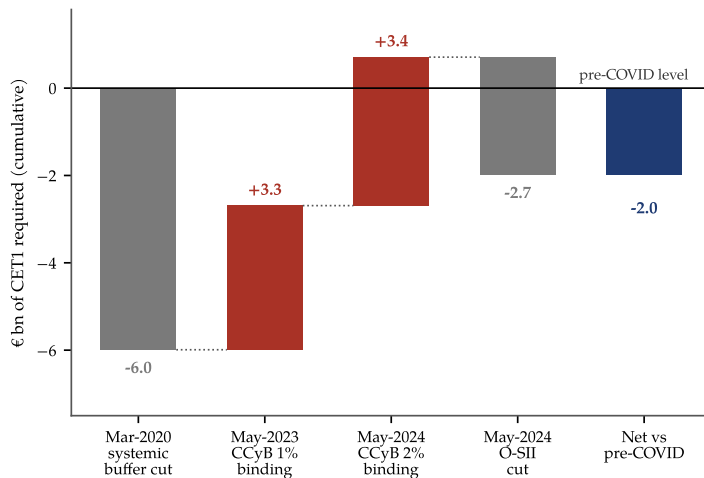
TERMS EXPLAINED

- **Phase 3 ("increased risk"):** The framework phase in which the CCyB goes above 2%.

NEXT → "And that last row tells us the refill was a swap, not a squeeze."

And the refill was a swap, not a squeeze: capital moved from structural buffers into the releasable one

Q2 · CRITERIA & RISKS



DNB's own words

"more or less capital-neutral for the three large banks" (2020)

"a limited increase in the net capital requirements" (Knot, Jun 2023)

"slightly increases the capital requirements for the banking sector as a whole" (FSR 2023)

Not unusual: the UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

Waterfall: approximate, own calculation. Buffer cuts valued at end-2022 total RWA of ING, Rabobank, ABN AMRO, de Volksbank (BNG excluded);

CCyB amounts are DNB's sector-wide figures on Dutch exposures. Relative to 2021 the change was an increase.

Source: DNB press release 17 Mar 2020; Knot (2023), statement to the Standing Committee for Finance, 7 Jun 2023; DNB FSR spring 2023; BCBS (2024) d585; bank annual reports 2022.

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SAY

- In euros, by our own estimate: the 2020 cut released about six billion; the CCyB then added 3.3 and 3.4 billion; the 2024 cut released another 2.7 billion.
- Net, compared with before COVID, the requirement is roughly unchanged. DNB called it "more or less capital-neutral".
- Two caveats: the bases differ, and compared with 2021 rather than 2019 it was a modest increase.
- So capital was **moved**, from buffers that cannot be released into one that can.
- Think of it as moving money from a locked savings account into a current account that the authority can open in a crisis. The total is similar; what changes is who can use it, and when.

NUMBERS

- -6.0 (2020: ING 1.7, Rabo 2.4, ABN 1.9) · $+3.3$ (May-23) · $+3.4$ (May-24) · -2.7 (May-24 O-SII) → net ≈ -2.0 €bn vs pre-COVID
- Valued at end-2022 total RWA; BNG excluded

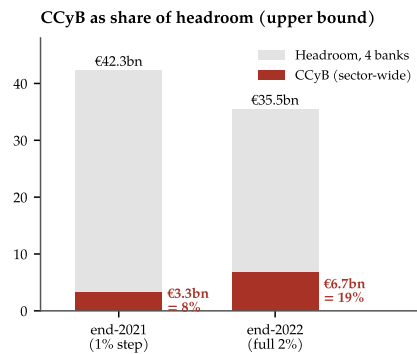
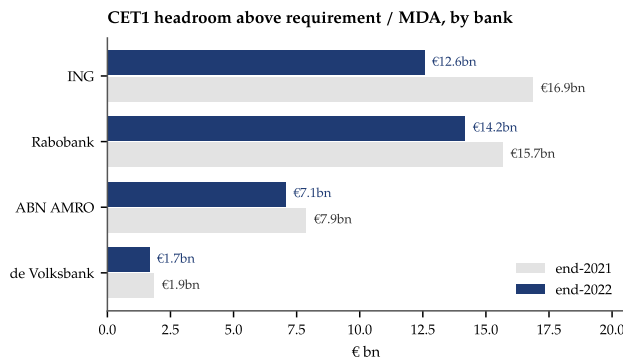
TERMS EXPLAINED

- **Structural buffer:** A capital charge for permanent features (size, interconnectedness), kept through the cycle.
- **UK approach:** The UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

NEXT → "So, was it the wrong moment?"

So was it the wrong moment? No. Higher rates gave banks the profits to build the buffer cheaply

Q3 · IMPLICATIONS



Headroom = CET1 capital above the bank's requirement (MDA trigger). Net interest income +6.7% in 2022 → buffers built from *retained earnings*, not by cutting loans. Headroom shrank because of RWA growth and payouts, not the CCyB.

ECB: gradual build-up + high profitability "limit these economic costs" (MPB 24); activating buffers during the 2022–23 tightening "can mitigate implementation costs" (MPB 31).

It was not the wrong moment. Arguably it was the right one.

Source: Bank disclosures (ING, Rabobank, ABN AMRO, de Volksbank, 2021–22), de Volksbank requirement derived; DNB FSR 2023; ECB Macropprudential Bulletin 24 (2024) and 31 (2025, Box 2).

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SAY

- No. The storm made it the **cheapest** moment.
- Headroom is capital above a bank's requirement. The four big banks had about 36 billion euros at the end of 2022; the full buffer of 6.7 billion is **at most a fifth** of that.
- Bank by bank: Rabobank had 14.2 billion of headroom, ING 12.6, ABN AMRO 7.1, and even the smallest, de Volksbank, 1.7 billion. The first step of 3.3 billion was only 8 percent of the total.
- And rising rates lifted profits, so banks could build the buffer from retained earnings instead of cutting loans.
- ECB research agrees: profitable banks and a gradual build-up keep the costs low, also during the 2022 to 2023 tightening.

NUMBERS

- Headroom end-2021 / end-2022 (€bn):
 - ING 16.9 / 12.6
 - Rabobank 15.7 / 14.2
 - ABN AMRO 7.9 / 7.1
 - de Volksbank 1.9 / 1.7
- Total headroom €42.3bn / €35.5bn → CCyB €3.3bn = 8%, €6.7bn = 19%

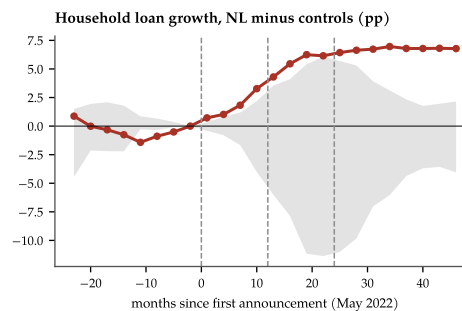
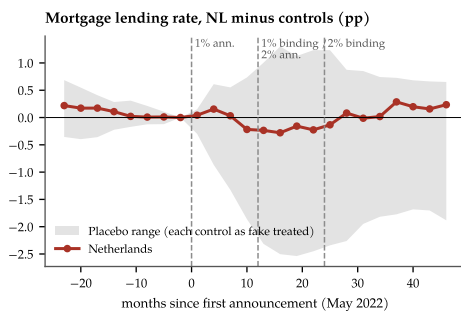
TERMS EXPLAINED

- **Headroom:** CET1 capital above the bank's requirement (its MDA trigger).
- **Retained earnings:** Profits kept in the bank instead of being paid out.

NEXT → "And borrowers did not pay either."

Borrowers did not pay either: credit kept flowing and its price moved with the rest of the euro area

Q3 · IMPLICATIONS



How to read: red = Netherlands minus countries that did not change their CCyB (AT, FI, IT, MT, LU); 0 = no difference; grey = differences that arise by chance (each control treated as if it had raised its CCyB).

- Dutch lending rates stayed **inside the placebo range** at all three CCyB dates
- Household credit grew **faster** than in peers (partly catch-up: NL lagged in 2019–22)

Source: ECB Data Portal (MIR, BSI); ESRB CCyB table (control selection); BIS (WS_TC); DNB press release 17 Sep 2026; own two-way fixed-effects event study.

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DNB (Sep 2026): “no signs of constraints in bank credit supply.”

One treated country, small control group, housing rebound: no sign of contraction, not a causal estimate.

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SAY

- We compare Dutch lending with five euro-area countries that never changed their buffer.
- The controls are Austria, Finland, Italy, Malta and Luxembourg. They kept their buffer unchanged throughout, so any difference is not driven by their own policy.
- The red line is the Netherlands minus those countries. The grey band shows the differences that arise by chance, when we pretend each control country was treated.
- Mortgage rates stay **inside the band** at all three dates. Household credit grew **faster**, partly as catch-up.
- In numbers: Dutch bank credit grew 16.6 percent from early 2022 to early 2026, against 7.9 percent in the euro area. Between 2019 and 2022 it had lagged, 4.3 against 11.3 percent. That is the catch-up.
- So: **no sign of contraction**, though not a causal estimate. DNB reached the same view in September 2026.

NUMBERS

- Dates: May-22 (1% announced) · May-23 (1% binding, 2% announced) · May-24 (2% binding)
- Bank credit 2022Q1–2026Q1: NL +16.6% vs euro area +7.9%; 2019–22: NL +4.3% vs +11.3%

TERMS EXPLAINED

- **Event study:** Tracks the difference between treated and control groups before and after an event.
- **Placebo test:** Rerun the analysis pretending an untreated country was treated, to see what chance alone produces.

NEXT → HAND-OVER: “So banks and borrowers were fine. What about the wider effects? Zheng.”

Nor did the buffer double the ECB's squeeze. It let monetary policy focus on inflation

Q3 · IMPLICATIONS

The concern

Two tightenings at once: ECB rates *and* capital requirements → credit crunch?

What the data show

Dutch lending rates did not diverge from euro-area peers (event study); bank profits *rose* with rates (net interest income +6.7%).

ECB Macprudential Bulletin 31 (2025)

Early CCyB activation "helps monetary policy focus on its primary objective of price stability, thereby largely eliminating the potential for conflict between price stability and financial stability."

DNB, FSR 2022

"A sudden tightening of financial conditions could also have a negative impact on financial stability."

One monetary policy for the euro area, but national macroprudential policy: the CCyB is the **Dutch-specific lever** and the insurance against a rate shock gone wrong.

Source: ECB Macprudential Bulletin 31 (Detken, Hempell & Pirovano, 2025); DNB Financial Stability Report, spring 2022.

Huang & Gutiérrez · The Netherlands' CCyB

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SAY

- Thank you. First, the buffer did not double the ECB's squeeze: Dutch lending rates moved with their peers, and bank profits rose with rates.
- The ECB argues that building buffers early lets monetary policy focus on inflation.
- With one monetary policy for twenty countries, the CCyB is the **Dutch-specific lever**, and insurance if a rate shock goes wrong.

NUMBERS

- ECB hiking from Jul 2022 (first +50bp) to at least Sep 2023

TERMS EXPLAINED

- **Macroprudential policy:** Policy aimed at the stability of the financial system as a whole.
- **Monetary policy:** Interest-rate policy aimed at price stability (ECB, euro area-wide).

NEXT → "Through reciprocity, the 2% also reaches foreign lenders."

Through reciprocity the 2% also binds foreign lenders, and the rest of Europe is moving the same way

Q3 · IMPLICATIONS

Across borders

- Foreign banks with Dutch exposures must also hold the 2% (mandatory reciprocity up to 2.5%)
- Limits leakage to foreign lenders, a known problem with national capital tools (Aiyar, Calomiris & Wieladek, 2014); *non-bank* leakage remains
- In April 2023, 5 banking-union countries had positive-neutral frameworks; NL had the **highest** rate (2%)

CCyB increases in the euro area

Country	Rate	Applies from
Germany	0.75%	Feb 2023
France	1.0%	Jan 2024
Ireland	1.5%	Jun 2024
Netherlands	2.0%	May 2024
Belgium	1.0%	Oct 2024
Portugal	0.75%	Jan 2026
Spain	1.0%	Oct 2026

Source: DNB press release 31 May 2023; ESRB CCyB rates table (Sep 2026); ECB Macropprudential Bulletin 21 (Behn et al., 2023); Aiyar, Calomiris & Wieladek (2014), JMCB.

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SAY

- Foreign banks lending into the Netherlands must hold the Dutch two percent too, which limits leakage to foreign lenders. Leakage to non-banks remains.
- And Europe is moving the same way: Germany, France, Ireland and Belgium raised their buffers in 2023 and 2024, and Portugal and Spain follow in 2026.

NUMBERS

- Germany 0.75% (Feb 2023) · France 1.0% (Jan 2024) · Ireland 1.5% (Jun 2024) · NL 2.0% (May 2024) · Belgium 1.0% (Oct 2024) · Portugal 0.75% (Jan 2026) · Spain 1.0% (Oct 2026)
- Positive-neutral rates in 2023: Cyprus 0.5%, Estonia and Lithuania 1.0%, Ireland 1.5%, NL 2.0%

TERMS EXPLAINED

- **Reciprocity:** Foreign banks apply the host country's CCyB rate to exposures there (mandatory in the EU up to 2.5%).
- **Leakage:** Credit shifting to lenders not covered by the rule (foreign branches, non-banks).

The result: next time, DNB can release €6.7bn overnight

Q3 · IMPLICATIONS

€0 → €6.7bn

releasable-by-design capital
2020 → 2024

≈ half

of the €12bn peak losses
after the GFC

€150bn

lending headroom
if released

Asymmetric payoff (Lang & Menno, 2023)**Normal times:** +1pp requirement → about **−0.1%** lending.**Bad times:** 1pp release → up to about **+10%** lending.

Cheap to hold, powerful to release.

A growing role

The mortgage risk-weight floor expires on 30 Nov 2026.

DNB: this “underpins the importance of . . . the current countercyclical capital buffer (CCyB) of 2%.”

Source: DNB framework (2022); Lang & Menno (2023), ECB WP 2828; DNB press release 24 Apr 2026.

Huang & Gutiérrez · The Netherlands' CCyB

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SAY

- In 2020 the Netherlands had no releasable capital. Today it has **6.7 billion euros**, about half of past peak losses and enough to support up to 150 billion of lending.
- It is cheap to hold and powerful to release: research suggests a release in bad times raises lending far more than the build-up costs in good times.
- In numbers: a one-point build-up in normal times reduces lending by about 0.1 percent, while a one-point release in bad times can raise it by up to ten percent.
- And with the mortgage floor expiring in November 2026, it matters even more.

NUMBERS

- €0 → €6.7bn releasable · ≈ half of €12bn peak losses · up to €150bn of lending
- Lang & Menno (2023): **−0.1%** lending per +1pp (normal times) vs up to **+10%** per 1pp release (bad times)

TERMS EXPLAINED

- **State-dependent effect:** The impact of a policy depends on the state of the economy (small in good times, large in bad times).

The price is reliance on discretion, and the buffer has never been tested

Q3 · IMPLICATIONS

Credibility	Without a mechanical rule, markets must trust DNB's judgement and communication. This is the classic rules-vs-discretion trade-off (Kydland & Prescott, 1977).
Calibration	Why 2%, not 1.5%? The ECB's loss-based range is 1.1–1.8%; NL chose the top of the banking union.
Untested	No release yet. Will banks lend the freed capital, or hoard it as they did with usable buffers in COVID?
Baseline	Relative to 2021 (not 2019) it <i>was</i> an increase: +€3.3bn before the O-SII offset.
Same profits, two claims	The ECB lists the Netherlands among countries adopting bank levies on excess profits in 2022–23. These partly claim the earnings that fund the buffer.

Source: Kydland & Prescott (1977); De Nora et al. (2025); ECB Macroprudential Bulletin 21 (2023) and 31 (2025); DNB (2022, 2023).

Huang & Gutiérrez · The Netherlands' CCyB

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SAY

- **Credibility:** without a rule, markets must trust DNB's judgement, so it has to explain itself every quarter.
- **Calibration:** why two percent, when ECB estimates suggest 1.1 to 1.8?
- **Untested:** will banks lend released capital, or hoard it?
- In 2020 many banks did not use the buffers they were allowed to use. A released CCyB is different, because the requirement itself falls, but that has not yet been tested in the Netherlands.
- **Baseline:** compared with 2021, it was an increase.
- **Profits:** the same bank profits were also targeted by excess-profit levies.

NUMBERS

- ECB loss-based range 1.1–1.8% vs NL 2% · +€3.3bn vs 2021 before the O-SII offset

TERMS EXPLAINED

- **Rules vs discretion:** The trade-off between predictable formulas and flexible judgement (Kydland & Prescott 1977).
- **Excess-profit levy:** A temporary tax on bank profits seen as windfalls from higher rates.

In short, the promise was kept: the buffer was refilled in a storm, at the moment it was cheapest

EPILOGUE

Q1 · Environment

Recovered economy hit by war, inflation and rate hikes; credit cold; housing hot, then cooling; strong, profitable banks.

Q2 · Criteria & risks

2% normal-times level, calibrated on past losses; insurance against unforeseeable shocks; housing to other tools; structural buffers cut in return.

Q3 · Implications

Cheap · largely a swap · no sign of contraction · complements monetary policy · reciprocated abroad · €6.7bn releasable · relies on discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it, even in a storm.

Huang & Gutiérrez · The Netherlands' CCyB

22/24

SAY

- **Environment:** a recovered economy in a storm, cold credit, cooling housing and strong banks.
- **Criteria and risks:** a two-percent normal level based on past losses, for shocks nobody can forecast, with housing left to other tools.
- **Implications:** cheap, largely a swap, no sign of contraction, and 6.7 billion euros now releasable, at the price of discretion.
- *(pause)* In March 2020 the Netherlands had nothing to release. **Next time, it will.** The best time to fill a buffer is when nobody thinks you need it. Thank you.

NUMBERS

- 0% → 2% · −33 / −47pp gap · $\leq 1/5$ of headroom · €6.7bn releasable

TERMS EXPLAINED

- **Normalisation:** Returning the buffer to its normal-times level, not reacting to overheating.

Glossary

APPENDIX

CCyB	Countercyclical capital buffer: 0–2.5% of risk-weighted assets on domestic exposures; raised with a 12-month lag, released immediately.	Credit-to-GDP gap	Credit/GDP ratio minus its long-run trend; the Basel reference for the CCyB (gap 2–10pp → buffer 0–2.5%).
Positive neutral CCyB	A CCyB kept above zero when risks are neither high nor low, so it can be released for any shock.	HP filter	Hodrick-Prescott filter: estimates a smooth trend; one-sided uses only past data (real time), two-sided also future data.
CET1	Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA.	Headroom	CET1 capital above a bank's requirement / MDA trigger.
RWA	Risk-weighted assets: assets weighted by their riskiness (e.g. mortgages count less than corporate loans).	Reciprocity	Foreign banks must apply the Dutch CCyB rate to their Dutch exposures.
Pillar 1 / P2R / P2G	Legal minimum (4.5% CET1) / bank-specific add-on set by the supervisor / non-binding supervisory guidance.	Borrower-based measures	Limits on loan-to-value (LTV) and loan-to-income (LTI) of new mortgages.
CCoB	Capital conservation buffer, 2.5% for all banks.	Risk-weight floor	Minimum average risk weight on Dutch mortgages of banks using internal models (Art. 458 CRR), Jan 2022 – Nov 2026.
O-SII / SRB	Buffer for other systemically important institutions / systemic risk buffer (structural, not designed for release).	Event study / placebo	Compare NL with countries that did not change their CCyB around each date / repeat with each control as a fake "treated" country.
MDA	Maximum distributable amount: if CET1 falls below the combined buffer requirement, payouts are automatically restricted.	pp / bp	Percentage point / basis point (0.01pp).

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SAY

- Backup only. Use it when a question involves a term; point to the row and give the one-line definition.

NUMBERS

- (definitions only)

TERMS EXPLAINED

- All key terms of the talk are on the slide

Methods and references

APPENDIX

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: (CET1 ratio – CET1 requirement/MDA) $\times$ total RWA per bank; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

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SAY

- Backup for method questions.
 - **Credit gap:** one-sided and two-sided HP filter ($\lambda = 400,000$), plus Hamilton regression filter.
 - **Headroom:** (CET1 ratio minus requirement) $\times$ total RWA, per bank.
 - **Event study:** two-way fixed effects, three-month bins, placebo inference over five never-treated countries.
- All references are verified.

NUMBERS

- Data: BIS, ECB Data Portal (MIR, BSI), ESRB, DNB, bank reports

TERMS EXPLAINED

- **Two-way fixed effects:** Regression with country and time dummies, so differences are measured relative to common trends.